Chapter 2

Preliminaries

2.1 Knots and diagrams 13
2.2 Crossing number and writhe 15
2.3 Knotation and not-tables 16
2.4 Seifert surfaces and genera 18
2.4.1 Special diagrams 21
2.5 Graphs 22
2.5.1 General definitions 22
2.5.2 The checkerboard coloring and checkerboard graph 24
2.6 Diagrammatic moves 25
2.6.1 Flypes and mutations 25
2.6.2 Bridges and wave moves 27
2.7 Braids and braid representations 28
2.8 Link polynomials 29
2.8.1 Generalities 29
2.8.2 Skein, Jones, and Alexander polynomial 30
2.8.3 Kauffman polynomial and semiadequacy 33
2.9 MWF inequality, Seifert graph and graph index 36
2.10 The signature 38
2.11 Genus generators 41
2.12 Knots vs. links 45

Taylor and Francis
Not for Distribution

2.1 Knots and diagrams

A *knot* will be an embedding of the circle S^1 in S^3 (or $\mathbb{R}^3$), a *link* an embedding of a collection of circles. These will be called *components* of the link. Thus a knot is a link of 1 component. (Usually the case of knots is included in a statement about links, but it will sometimes depend a bit on the context of the statement so a little care may be necessary.) The *number of components* of a link L will be written as $n(L)$ (so $n(K) = 1$ if K is a knot). We will assume $n(L) \geq 1$, i.e., we will not treat the empty link. To avoid wild behavior, it suffices to assume, e.g., that knots and links are smooth and regarded up to smooth isotopy of the ambient space.

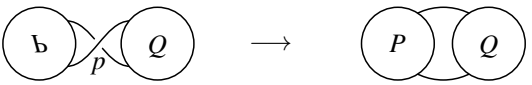
Knots and links are commonly represented by *diagrams*, which consist of smooth plane curves with *crossings*. Crossings are understood to be transverse intersections

© 2016 by Taylor & Francis Group, LLC

(self-intersections of a curve or intersections of different curves for links) with added over- and undercrossing strand information.

Let D be a diagram of a knot of link L. We write also $n(D) = n(L)$ for the *number of components* of D, and $c(D)$ for the *number of crossings* of D.

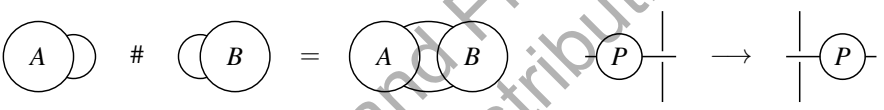
A crossing p in a link diagram D is called *reducible* (or nugatory) if it looks like the left side of Figure 2.1. We call D reducible if it has a reducible crossing; otherwise we call it *reduced*. The reducing of the reducible crossing p is the move depicted in Figure 2.1. Each diagram D can be (made) reduced by a finite number of these moves.



(2.1)

FIGURE 2.1: Eliminating a reducible crossing

In the following we assume that all diagrams are reduced unless otherwise stated.



(a) (b)

FIGURE 2.2: Diagram connected sum and factor slide move.

Part (a) of Figure 2.2 displays the *connected sum* $D = A\#B$ of the diagrams A and B. The latter are called *factors* of D. The connected sum of diagrams is uniquely determined only up to the move shown in part (b) of the figure (and the mirror image of that move). We will call it a *factor slide*. If a diagram D can be represented as the connected sum of diagrams A and B, such that both A and B have at least one crossing, then D is called *composite*; otherwise it is called *prime*. A knot of link L is *prime* if whenever $D = A\#B$ is a composite diagram of L, one of A and B represent an unknotted arc (but not both; the unknot is not prime per convention).

A diagram D is *connected* if its curve is a connected set in $\mathbb{R}^2$, that is, there is no closed curve γ disjoint from D such that both the interior and exterior of γ have non-trivial intersection with D. Otherwise D is *disconnected* or *split*. A link is *split* if it has a split diagram and *non-split* if otherwise.

One of the classic primality criteria, for alternating diagrams, is due to Menasco.

Theorem 2.1.1 ([Me]) If D is a prime alternating non-trivial diagram of a link L, then L is prime.

Downloaded by [CRCNetbase T&F Offices] at 03:10 10 August 2016

Taylor and Francis
Not for Distribution

© 2016 by Taylor & Francis Group, LLC

2.2 Crossing number and writhe

The *crossing number* $c(K)$ of a knot K is the minimal crossing number $c(D)$ of all diagrams D of K.

This is a very basic quantity, but it is incredibly hard to determine in many situations and plagues many knot theorists. It is suggestive that crossing number is additive under connected sum,

$$c(K_1 \# K_2) = c(K_1) + c(K_2). \tag{2.2}$$

This is known for special classes (e.g., alternating knots), and a computer can confirm it for a myriad of instances. But the problem is open (and difficult) in general, and one among the many worthwhile and important problems too far afield from the focus of this text.

The *writhe*, or *(skein) sign*, is a number ± 1, assigned to any crossing in a link diagram. A crossing as in Figure 2.3(a) has writhe 1 and is called *positive*. A crossing as in Figure 2.3(b) has writhe -1 and is called *negative*. The *writhe* $w(D)$ of a link diagram D is the sum of writhes of all its crossings.

Let $c_\pm(D)$ be the number of positive, respectively negative crossings of a diagram D, so that $c(D) = c_+(D) + c_-(D)$ and $w(D) = c_+(D) - c_-(D)$. Let $c_\pm(K)$ for a knot K denote the minimal number of positive respectively negative crossings of a diagram of K.



FIGURE 2.3: A positive, negative, and smoothed out crossing.

A diagram is *positive* if all its crossings are positive. It is *almost positive* if all of its crossings are positive except for exactly one. A knot or link is *positive* if it has a positive diagram. See, e.g., [CM, O, Yo, Zu][1]. It is *almost positive* if it is not positive but has an almost positive diagram. More generally, a diagram D is *k-almost positive* if it has exactly k negative crossings, i.e., $c_-(D) = k$, and a knot K is k-almost positive if it has a k-almost positive but no $k-1$-almost positive diagram, i.e., $c_-(K) = k$. The switch of crossings of a diagram D so that all become positive will sometimes be called *positification* of D. Analogously one can define *negative* diagrams and knots.

A diagram is *almost alternating* [Ad, Ad3, Ad2, GHY] if it can be turned by one crossing change into an alternating one. A knot is *almost alternating* if it has an almost alternating diagram but is not alternating.

[1] Beware that some authors use the name for the narrower class of links with positive braid representations.

Downloaded by [CRCNetbase T&F Offices] at 03:10 10 August 2016

Taylor and Francis
Not for Distribution

© 2016 by Taylor & Francis Group, LLC

We write $!D$ for the *mirror image* of D. We obtain $!D$ by switching all crossings of D. Similarly $!K$ denotes the mirror image of K (i.e., $!K$ has the diagram $!D$, when K has the diagram D). Clearly $g(!D) = g(D)$ (and therefore $\tilde{g}(K) = \tilde{g}(!K)$) and $g(!K) = g(K)$.

Below we consider *amphicheiral*, or *achiral* knots. This is the designation of a knot which can be turned into its mirror image: $K = !K$. From the knots in Figure 1.1, the unknot is obviously amphicheiral. So is the figure-eight knot, as shows a simple exercise. In contrast, the trefoil is not amphicheiral. In other words, the left-hand trefoil and its mirror image, the right-hand trefoil on Figure 2.4, are two distinct knots. A knot like the trefoil which is not amphicheiral is called *chiral*.

2.3 Knotation and not-tables

Apart from their historic role, knot tables serve, also for us below, as an indispensable source for examples and counterexamples. In practice, it turns out useful to tabulate, for a given crossing number, only prime knots. (Note that this release from duty bears to some extent the ascertaining of (2.2).) Further, one usually disregards orientation of the knot and treats each knot only up to taking the mirror image (i.e., among a knot and its mirror image, only one is included).

Nowadays a prime knot is typically denoted as X_Y, where X is the crossing number, and Y is the order in the table. For example, there is only one (prime) knot of crossing number 3 (up to mirror image), the trefoil, and consequently it will be written 3_1. Similarly 4_1 is the figure-8-knot. But for 5 or more crossings, there is more than one prime knot, so the question is: What is the order? However commendable many tabulators' efforts, it seems also necessary to recount some of the pitiful history on this very basic issue.

According to the goal of a table, knots included therein more than once (duplications) or not at all (omissions) constitute errors. The compilation (notably in the early stages) and presentation of these tables has propagated not only such errors, of which the Perko pair [Pe] is likely only the most notorious one, but even errors in the errors [E]. The origin of this particular plague is some discord over the last four knots in Rolfsen's tables [Ro, appendix] after Perko's duplication emerged. Their indices are often shifted down by 1. Some record on this and other problems in the table (like the mix-up of 10_{83} and 10_{86}) can be found on Bar-Natan's Knot Atlas [BN].

Even if correct, different compilation procedures will, of course, render the list in a different order. Apparently, the attempts to popularize one's own (or, at least, the failure to coordinate each other's) ordering, and to disregard previous conventions, have a tradition almost as long as tabulation itself. Let me pay tribute to a senior who should know better and whose discussion [Pe3] is unpublished, but I find worth hearing.

Taylor and Francis
Not for Distribution

Downloaded by [CRCNetbase T&F Offices] at 03:10 10 August 2016

© 2016 by Taylor & Francis Group, LLC

> Note that Reidemeister [Re], following Alexander and Briggs [AB], does not use Tait's original numbering [T]—their 9_{23} is Tait's 9-XII. So Bar-Natan was not the first Princeton-educated topologist to tamper[2] with the previously well established numbering of tabled knots (see [E] and [St22] for consequences). Others have also chosen to ignore the historic nomenclature handed down by Tait, Little, and Conway [Ro, HTW] and, alas, our own efforts to preserve the established order in papers classifying the 10- and 11-crossing knots [Pe, Pe4] have long been consigned to the dust bin of history, most recently under the rubric of "old Perko's notation" [CDM, p. 49].[3]

Rolfsen's tables of ≤ 10 crossing knots (which accord with Reidemeister and Alexander-Briggs) are now widely accepted at least because of the importance his book has gained. We thus feel compelled to stick to this quasi-standard here (and are lucky to mostly avoid its described problems).

For *knots from 11 to 16 crossings* our numbering is that of the tables *included in KnotScape* [HT]. This choice is simply because I largely appealed to KnotScape's functionality for diverse computations. However, I reorganize the KnotScape tables so that non-alternating knots are appended after alternating ones (of the same crossing number), instead of using "a" and "n" super/subscripts. For example, there are 367 prime alternating knots of 11 crossings. Thus a knot commonly named 11_{a111} in the table will be written as 11_{111}, while 11_{n111} as 11_{478} (for $111 + 367 = 478$).

Knot diagrams of 17 and 18 crossings were generated by the algorithm (and software) of [RFS], and are applied for some computational checks, but a numbering is never used.

As a noteworthy sequel to the above story of Perko, be aware that Hoste–Thistlethwaite break even with the Rolfsen numbering for ≤ 10 crossing knots, and they do so without clear warning. (Already the two 5 crossing knots are exchanged.) Visible bruises of the obstinacy protrude on [Kf4, p. 218] (where 10_{48} is written as 10_{79}) and in the knot tables of the very first edition of Adams' Knot book [Ad]. For my own use, I manually recoded a part of KnotScape; I saw this as the least awkward way of preventing more annoying miscommunication. See [An] for an entire book on tabulation yet completely independent from [HTW] and previous efforts.

As a further small twist, it will be necessary for us to distinguish at a few places below between a knot and its mirror image. We thus have to fix which knot should be X_Y and which $!X_Y$. We will use the *mirroring convention in KnotScape's tables throughout*. It is not easy to describe in a few words, but one distinctive feature is that all diagrams there have at least one positive crossing. This explains the choice for all

[2] To set the record straight: Bar-Natan merely clarifies and follows a practice (regarding the last four Rolfsen knots) initiated long before, notably in [BZ], and reiterated since by many others (incl. myself).

[3] Besides all his valid criticism, one should also take into account that Perko, as a hobby-mathematician (and professional lawyer), achieved a publishing record enviable to many of us so-called mathematics "professionals" (incl. myself).

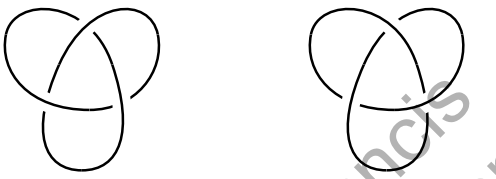
Downloaded by [CRCNetbase T&F Offices] at 03:10 10 August 2016

Taylor and Francis
Not for Distribution

© 2016 by Taylor & Francis Group, LLC

knots occurring there with positive diagrams, which includes all special alternating knots, and should suffice here.

Our use of mirroring for ≤ 10 crossing knots means that a knot X_Y in Rolfsen's tables will be mirrored for us the way it occurs in KnotScape's tables. Most importantly, 3_1 will be thus the positive (right-hand) trefoil, while $!3_1$ the negative (left-hand) one—*unlike* in Rolfsen's table. Keep in mind that the knot usually appears in KnotScape at a different place. Thus, for instance, 5_1 will be for us the $(2,5)$-torus knot mirrored positively, even although it is mirrored negatively by Rolfsen, and written as 5_2 by Hoste–Thistlethwaite. A detailed conversion list can be found on my website [St23].



left-hand trefoil $!3_1$ right-hand trefoil 3_1

FIGURE 2.4: The two trefoils

Caution—and, desirably, explanation—of what one uses and how thus becomes a to-be-present—even if not pleasant—companion of taking advantage of knot tables. (The case of links is a Pandora's box I do not even dare to open…) I personally find history deserves some care to be maintained, even by champions, rather than the amount of confusion, hassle, and dismay that results from taking it less seriously. Perko finishes his note by quoting Reidemeister [Re2, p. 92]: "Das mathematische Denken is nur der Anfang des Denkens."[4]

2.4 Seifert surfaces and genera

A *Seifert surface* of an oriented knot or link L is a compact oriented surface S embedded in S^3 with boundary $\partial S = L$.

$$S \hookrightarrow S^3, \qquad \partial S = L. \tag{2.3}$$

Whenever S is oriented, there is an induced orientation on $\partial S = L$, and we will assume that this orientation coincides with the given orientation of L for each component of

[4] "Mathematical thinking is only the beginning of thinking."

Downloaded by [CRCNetbase T&F Offices] at 03:10 10 August 2016

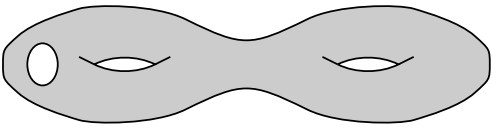
Taylor and Francis Not for Distribution

© 2016 by Taylor & Francis Group, LLC

L. It is possible that S is disconnected (although these cases will not be of much interest here; see Theorem 2.8.1 for the reason). However, we will assume that no component of S is closed, that is, each component of S bounds at least one component of L.

By the classification of compact oriented surfaces, a connected surface is determined by its *genus* $g = g(S)$ (pictorially, the number of holes of S) and the number of *boundary components* n.

The example for $n = 1$ and $g = 2$ is:

$S =$  (2.4)

The *Euler characteristic* of S is

$$\chi(S) = 2 - 2g - n. \tag{2.5}$$

The *(Seifert) genus* $g(L)$ of a knot or link L is given by

$$g(L) = \min\{\, g(S) : S \text{ is a Seifert surface of } L \,\}.$$

The unknot is the only knot of genus 0. Similarly, the *Euler characteristic* $\chi(L)$ is said to be the maximal Euler characteristic of Seifert surface of L. It is immediately clear from (2.5) that when L bounds no disconnected Seifert surface, then

$$2g(L) = 2 - n(L) - \chi(L).$$

For a diagram D of L, there is a procedure of Seifert [Se], called *Seifert algorithm*, to construct such a surface whenever a diagram D of L is given. The procedure is briefly this.

The replacement of a positive or negative crossing (a) respectively (b) in Figure 2.3 by the no-crossing fragment (c) appearing on the right will be called *smoothing out* the crossing. First, smooth out all crossings of D (but keep information on where crossings were located). The resulting disjoint plane loops are called the *Seifert circles* of D. Then, plug disks into all Seifert circles, and finally connect these disks by half-twisted bands to reinstall the crossings (at the place they were originally smoothed out).

After applying this procedure, we now observe the knot or link depicted in D to carry a surface, which is easily seen to be orientable (and respecting the orientation of L). Such a Seifert surface will be called *canonical* (for the diagram D) and will play a major role below.

The *(canonical) genus* $g(D)$ of D is defined to be the genus of the Seifert surface obtained by Seifert's algorithm on D, and its Euler characteristic $\chi(D)$ is called *(canonical) Euler characteristic* (of D).

Downloaded by [CRCNetbase T&F Offices] at 03:10 10 August 2016

Taylor and Francis
Not for Distribution

© 2016 by Taylor & Francis Group, LLC

It is obvious that for every diagram D of L,

$$g(D) \geq g(L), \quad \text{and} \quad \chi(D) \leq \chi(L).$$

If equality holds, we will say that D has *minimal genus*. It should be noted that L possesses a multitude of Seifert surfaces, and not all of them occur as canonical surfaces for some diagram of L. Morton [Mo] has found the first examples of knots where in fact no canonical Seifert surface has minimal genus.

It should be clear that the embedding (2.3) will hardly be as neat as in (2.4). The surface in (2.4) can be set into S^3 so that its boundary realizes any genus 2 knot. Theorem 2.11.11 will reveal a multitude of knots where such surfaces are canonical, and there are many more with non-canonical ones. This serve as a caveat to an unexperienced reader against pre-conceived notions about the appearance of Seifert surfaces.

Note also that a surface S as in (2.3) is obviously not of minimal genus because "a handle can be cut off." This can be made more precise by saying that there is a loop $C \subset S$ bounding a disk in the complement of S that is not isotopable onto S. Such surfaces are called *compressible* and the others *incompressible*. A minimal genus surface must obviously be incompressible, but beware that the converse is not always true.

Write $s(D)$ for the *number of Seifert circles* of D. Then, with $c(D)$ the number of crossings, and $n(D) = n(L)$ the number of components as in §2.1,2.2, we have

$$\chi(D) = s(D) - c(D) \quad \text{and} \quad 2g(D) = 2 - n(D) - \chi(D).$$

Theorem 2.4.1 (see [Mu2, Cw, Ga, Cr]) If D has an alternating or positive diagram of L, then $g(L) = g(D)$ and $\chi(L) = \chi(D)$.

The *canonical genus* $\tilde{g}(L)$ respectively *canonical Euler characteristic* $\tilde{\chi}(L)$ is defined as the minimal genus respectively maximal Euler characteristic of all diagrams of L. As cautioned by Morton's examples (see [Mo]), in general we can have $g(L) < \tilde{g}(L)$.

Bennequin's inequality (theorem 3 in [Be]) can be stated, using (as explained in [St]) the work by Vogel [Vo], in the following way.

Theorem 2.4.2 If D is a diagram of a knot K, then $g(K) \geq g(D) - c_-(D)$.

In particular, if D is a diagram of the unknot, then $c_-(D) \geq g(D)$.

We call a crossing p connected to a Seifert circle s also *adjacent* or *attached* to s. The *valency* of a Seifert circle s is the number of crossings attached to s. We call a Seifert circle *negative* if only negative crossings are attached to it. Let $s_-(D)$ be the number of negative Seifert circles of D.

Bennequin's inequality was improved by Rudolph [Ru]. We state this improvement only for knots. Because a knot K lives in S^3, which bounds the 4-ball B^4, there are

Downloaded by [CRCNetbase T&F Offices] at 03:10 10 August 2016

Taylor and Francis
Not for Distribution

© 2016 by Taylor & Francis Group, LLC

(smooth) surfaces bounding K in B^4. The *smooth slice genus* $g_s(K)$ of K is the minimal genus of such a surface. Every Seifert surface can be pushed into B^4, and thus

$$g(K) \geq g_s(K)$$

for every knot K. A knot K is *(smoothly) slice* if $g_s(K) = 0$.

Theorem 2.4.3 If D is a diagram of a knot K, then

$$g(K) \geq g_s(K) \geq g(D) - c_-(D) + s_-(D)\,. \tag{2.6}$$

We will refer to (2.6) as the *Rudolph–Bennequin inequality*.

The notion of Seifert surfaces is fundamental, and every (serious) general knot theory textbook should cover it in some way, as well as at least also mentioning Seifert's method. The topology of Seifert surfaces is discussed at length in [Ro]. For a more combinatorial point of view, which is more useful here, see [Cr2, Chapters 2 and 5]. A basic treatise can also be found in Adams' book [Ad].

2.4.1 Special diagrams

A diagram is *special* if no Seifert circle contains other Seifert circles in both regions that it separates the plane into. Such Seifert circles are called *separating*.

Let us here for the sake of thoroughness briefly outline the reason for the following well-known but useful fact.

Lemma 2.4.4 For a connected diagram D, two of the properties special, positive/negative, and alternating imply the third.

Proof. We will show that a positive connected diagram D is alternating if and only if it is special. This argument will make clear how to obtain the rest of the statement.

Thus let D be connected and positive.

First, assume D has a separating Seifert circle s, i.e., D is not special. Because D is connected, there are crossings attached to s from inside and from outside. Let p, q be two consecutive crossings in cyclic ordering along s, such that p is attached to s from inside and q from outside. Then the overpassing strand of p is also an overpassing strand of q, and thus D is not alternating.

Now, conversely assume D is special. Assume a strand passes a crossing p from below. Then it becomes part of a Seifert circle s to which p is attached. The crossing q next to p in cyclic ordering along s is attached to s from the same side as p (because D is special and s is not separating). Thus the underpassing strand of p exits s as an overpassing strand of q. Consequently, every underpass in D is followed by an overpass, and one easily sees also that every overpass is followed by an underpass. Thus D is alternating. □

Taylor and Francis
Not for Distribution

Downloaded by [CRCNetbase T&F Offices] at 03:10 10 August 2016

© 2016 by Taylor & Francis Group, LLC

A diagram with the three properties in Lemma 2.4.4 is called *special alternating*. A knot is special alternating if it has a special alternating diagram. Such knots were introduced and studied by Murasugi [Mu] and have a series of special features. Conversely all knots have a special (not necessarily alternating) diagram. Hirasawa [Hi2] shows how to a modify any knot diagram D into a special diagram D' so that $g(D) = g(D')$. (Actually, the canonical surfaces of D and D' are isotopic.) We will study his procedure in great detail below (§3.2).

The *index* $\mathrm{ind}(s)$ of a separating Seifert circle s is defined as follows: Denote for an inner crossing (i.e., attached from the inside) of s a letter i, and for an outer crossing a letter o cyclically along s. Then $\mathrm{ind}(s)$ is by definition the minimal number of disjoint subwords of the from i^n or o^n $(n > 0)$ of this cyclic word. (Put differently, it is the number of pairs p, q as in the first part of the proof of Lemma 2.4.4.) For example, the Seifert circle s in Figure 3.1 on page 49 has index 4.

2.5 Graphs

2.5.1 General definitions

There are some close connections between knot and link diagrams and graphs, which will also be very relevant here. One such is the Seifert graph (see §2.9). We must thus spend effort very early on to give some basic preliminaries on graphs.

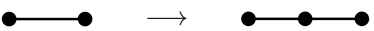
Note: Seifert graphs are one important occurrence of graphs in knot theory but not the only one by far. Another is the checkerboard graph, which we will use (§7.5) and thus discuss (§2.10) more briefly. For other graph-theoretic excourses, see, e.g., [CDM]. In §10.6 we will sketch some possible connection between the occurrence of trivalent graphs in Vassiliev invariant theory and our context.

A graph G will have for us possibly multiple edges (edges connecting the same two vertices) but usually no loop edges (edges connecting one and the same vertex). Edges connecting the same two vertices will be called *parallel*.

By $V(G)$ we will denote the set of vertices of G, and by $E(G)$ the set of edges of G, with each multiple edge counting as a set of single parallel edges; $v(G)$ and $e(G)$ will be the number of vertices and edges of G (thus counted), respectively.

The number of edges with endpoint $v \in V(G)$ is called the *valence* $\mathrm{val}_G(v)$ of v. A vertex of valence 0 is called *isolated*.

For a graph, let the operation



(adding a vertex of valence 2) be called *bisecting* and its inverse (removing such a

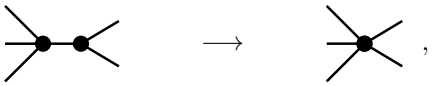
Taylor and Francis
Not for Distribution

Downloaded by [CRCNetbase T&F Offices] at 03:10 10 August 2016

© 2016 by Taylor & Francis Group, LLC

vertex) *unbisecting* (of an edge). If G has no vertices of valence 2, we call a graph G' a *bisection* of G if G' is obtained from G by a sequence of edge bisections. We call a bisection G' *reduced* if it has no adjacent vertices of valence 2. (For us usually G will be a trivalent graph, and a reduced bisection G' of G means that each edge of G is bisected at most once to obtain G'.) Conversely if G' is a graph, its *unbisected graph* G is the graph with no valence-2 vertices, of which G' is a bisection.

Similarly, a *contraction* is the operation



and a *decontraction* its inverse.

The *doubling* of an edge consists in adding a new (parallel) edge connecting the same two vertices.

A graph G is *series-parallel* if it can be obtained from  by edge iterated bisections and doublings.

A graph G is *connected* if for any two vertices $v, w \in V(E)$ there is a *path* connecting them, i.e., there are vertices $v_0 = v, v_1, \ldots, v_n = w$ with an edge between v_i and v_{i+1} in G. A graph which is not connected is *disconnected*.

A *cycle* in a graph should be defined in our situation with a bit of caution to allow the following three options: it is (a) a single loop edge (will usually not occur), (b) a collection of two parallel edges, (c) a set of $n \geq 3$ edges e_i, $i = 1, \ldots, n$ such that there are vertices $v_1, \ldots, v_n$ of G with e_i connecting v_i and $v_{i \bmod n+1}$.

A graph is a *forest* if it has no cycles. It is a *tree* if it has no cycles and is connected. (According to this description, an isolated vertex is a tree.)

A graph is *n-connected* if n is the minimal number of edges needed to remove from it to disconnect it. (Thus connected means not 0-connected.) Such a collection of edges is called an *n-cut*.

From now when we delete an edge, we understand that a vertex it is incident to is *not* to be deleted too. In that sense, the set $\operatorname{star} v$ of edges incident to a given vertex v, called *star* of v, always forms a cut—if we delete these edges, v becomes an isolated component; thus the graph is not connected anymore. A *cut vertex* is a vertex which disconnects a graph when removed *together* with all its incident edges.

A graph G is *planar* if it is embeddable in the plane *and equipped* with a fixed such planar embedding. Observe that there is a natural bijection of edges between a planar graph G and its *dual graph* G^*; in that sense we can talk of the dual $e^* \in E(G^*)$ of an edge $e \in E(G)$. The operations doubling and bisecting become dual to each other.

A *subgraph* Γ of G is a graph Γ with $E(\Gamma) \subset E(G)$ and $V(\Gamma) = V(G)$. (Thus, according to this definition, we may have $\operatorname{val}_\Gamma(v) = 0$ while $\operatorname{val}_G(v) > 0$.) A graph Γ is a *spanning tree* of G if Γ is a subgraph of G and Γ is a tree.

Taylor and Francis
Not for Distribution

Downloaded by [CRCNetbase T&F Offices] at 03:10 10 August 2016

© 2016 by Taylor & Francis Group, LLC

A graph G is *bipartite* if there is a disjoint division $V(G) = V_1 \cup V_2$ (with $V_1 \cap V_2 = \varnothing$), such that each edge in G connects $v \in V_1$ with $w \in V_2$.

2.5.2 The checkerboard coloring and checkerboard graph

Definition 2.5.1 A *region* of a link diagram is a connected component of the complement of the (plane curve of) the diagram. An *edge* of D is the part of the plane curve of D between two crossings (clearly each edge bounds two regions). At each crossing p, exactly two of the four adjacent regions contain a part of the Seifert circles near p. We call these the *Seifert circle regions* of p. The other two regions are called the *non-Seifert circle regions* of p. If the diagram is special, each Seifert circle coincides with (the boundary of) some region. We call the regions accordingly Seifert circle regions or non-Seifert circle regions (without regard to a particular crossing).

Note that a connected diagram of n crossings has $n+2$ regions.

To any link diagram D one can associate its *checkerboard graph* $G = G(D)$ in the following manner. Consider the plane curve $\hat{D}$ of D as a 4-valent planar graph.

A *checkerboard coloring* of a diagram D is obtained as follows. We assign to regions (connected components of the complement) of the diagram D in the plane colors black and white so that regions bounded by a common edge of D have opposite color. Let $B(D)$ be the number of black and $W(D)$ of white regions. The total number of regions is

$$W(D) + B(D) = c(D) + 2\,. \tag{2.7}$$

Then $G(D)$ is defined to have a vertex for each black region of $\hat{D}$ and an edge for each crossing. See Figure 2.5 (or also, e.g., [Th4, Kf2]). We should note that $G(D)$ is in fact a *planar* graph, i.e. comes with a planar embedding.

To avoid pathologies occurring with $G(D)$, we will assume from now on that a link diagram D is connected and hence so is $G(D)$. This is automatically true for knot diagrams, but in the few situations where we discuss links, we will consequently assume that links are non-split.

When D is connected, then a switch between color in the checkerboard coloring of D results in taking the dual of the checkerboard graph. This graph is thus defined up to duality.

A diagram D is *arborescent* if its checkerboard graphs are series-parallel (see e.g., [St17]); it is easy to see that the property series-parallel is preserved under duality. A knot or link is arborescent (or Conway-algebraic[5]) if it has an arborescent diagram.

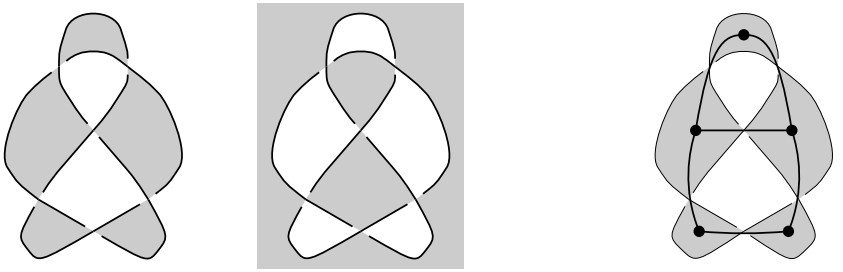
The class of arborescent links contains the *Montesinos links*, which in turn contain the *2-bridge (or rational)* links. These classes are a little more difficult to describe in the present setting and are better characterized in terms of their Conway notation

[5] Beware that there is another, and totally different, class of knots named "algebraic"!

Downloaded by [CRCNetbase T&F Offices] at 03:10 10 August 2016

Taylor and Francis
Not for Distribution

© 2016 by Taylor & Francis Group, LLC



two checkerboard colorings

the checkerboard graph of the first coloring

FIGURE 2.5: Checkerboard coloring and checkerboard graph.

[Co]. A good account on Conway's notation (at least for arborescent links) is given in [Ad, §2.3]. For a more advanced treatment of Montesinos and 2-bridge links, see [BZ]. Montesinos and 2-bridge links will play a comparatively minor role below, and thus we prefer to leave the reader with these references instead of opening yet another thread here.

Taylor and Francis
Not for Distribution

2.6 Diagrammatic moves

Moves between link diagrams, not necessarily preserving the underlying link, will play an important role. We already learned about those of Reidemeister in §1.2. Here we discuss further moves which will be needed later. (This list is not complete, one major addition being the $\vec{t}_2'$ twist introduced in §2.11.)

2.6.1 Flypes and mutations

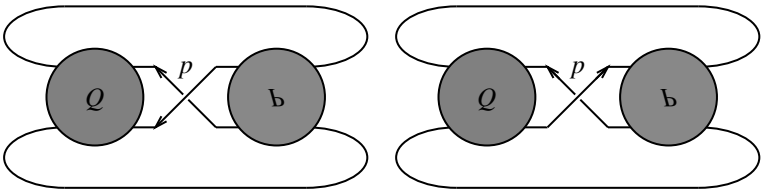
Definition 2.6.1 A *flype* is a move on a diagram shown in Figure 1.2. We say that a crossing *admits* a flype if it can be represented as the distinguished crossing in the diagrams in the figure and both tangles have at least one crossing.

By the fundamental work of Menasco–Thistlethwaite, we have a proof of the Tait Flyping conjecture.

Theorem 2.6.2 ([MT]) For two alternating diagrams of the same prime alternating link, there is a sequence of flypes taking the one diagram into the other.

Downloaded by [CRCNetbase T&F Offices] at 03:10 10 August 2016

© 2016 by Taylor & Francis Group, LLC

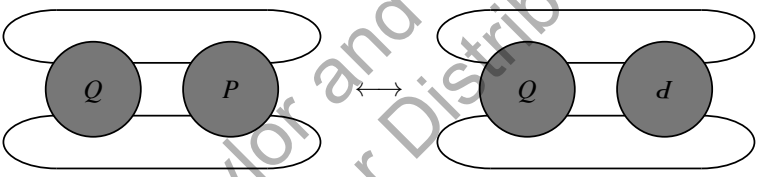


type A type B

FIGURE 2.6: A flype of type A and B.

We introduced (see [SV]) a distinction of flypes according to the orientation near the crossing p at which the flype is performed. See Figure 2.6. An important observation is that each crossing admits at most one of the types A and B of flypes, and this remains so after applying any sequence of flypes on the diagram. We will elaborate more on this point in §2.11.

Mutation was introduced by Conway [Co]. It is a procedure of rotating by π a 2-string tangle in a tangle decomposition of a link diagram D along the axis perpendicular to the projection plane or horizontal or vertical in the projection plane. For example:



(2.8)

(In oriented diagrams D, one may have to change the direction of all strands of either P or Q to make in- and outputs compatible.) The sphere, intersecting the link in 4 points, which separates both tangles, is called *Conway sphere*.

Knots and links are *mutants* if they are transformable into each other by a sequence of mutations. Mutations are very similar to flypes, and in fact one easily sees that one can realize a flype by a sequence of mutations. However, in general mutations do change the link type. Nevertheless, mutants are "similar" and difficult to distinguish because most of the common (efficiently computable) invariants coincide on them, including the polynomial invariants we will introduce below. This failure to distinguish mutants, although disappointing at first, can be utilized to reduce a lot of work at several places (see §2.11). Mutations start for knots at 11 crossings, and the (historically) most famous example are the Kinoshita–Terasaka and Conway knots (see, e.g., [CM2, LM, MR]).

Menasco's work [Me] on Theorem 2.1.1 also shows that Conway spheres in alternating link complements are visible in an alternating diagram with a small family of exceptions, the so-called Menasco's *hidden Conway spheres*. One consequence worth mentioning is that if an alternating link has a mutant, that mutant is also alternating, and the mutation is visible in alternating diagrams. In particular, *the unknot*

Downloaded by [CRCNetbase T&F Offices] at 03:10 10 August 2016

Taylor and Francis
Not for Distribution

© 2016 by Taylor & Francis Group, LLC

has no mutants, that is, a mutation applied on a diagram of the unknot will produce the unknot again (a circumstance known earlier). Lickorish's book [L, Chapter 4] contains more details.

2.6.2 Bridges and wave moves

A *bridge* is a piece of a strand of a knot diagram that passes only crossings from above. The *length* of the bridge is the number of crossings passed by it (*excluding* the initiating and terminating underpass). A *tunnel* is the mirror image of a bridge.[6]

A *wave move* is a replacement of a bridge a of length l_1 by another one b of length l_2. We will assume throughout that $l_2 \leq l_1$. The move is (crossing number) *reducing* if $l_2 < l_1$ (see Figure 2.7 or [StK] for example) and (crossing number) *preserving* if $l_2 = l_1$. Elsewhere a wave move is also called a (l_1, l_2)-*pass*. Note that a $(1,0)$-pass is the removal of a nugatory crossing (much like in Figure 2.1, except that P is flipped), and a $(1,1)$-pass is exactly the factor slide of Figure 2.2 (b). We will usually consider only reducing wave moves, except for the case $l_1 = l_2 = 1$. Obviously, such a move works for tunnels instead of bridges and we *will not distinguish between both* unless clearly stated.



FIGURE 2.7: Wave moves. The number of strands on left and right of the shaded circle may vary. It is only important that the parities are equal respectively different, and that the left-outgoing strands are fewer that the right-outgoing ones.

Adams introduced a *tongue move* allowing us to create more complicated almost alternating diagrams from a given one. This move is shown in Figure 2.8. Herein the crossing needed to be switched to obtain an alternating diagram is encircled; we call this crossing *dealternator*.

Adams formulated a conjecture, stating how to recognize the unknot in almost alternating diagrams, which was proved recently by Tsukamoto.

Theorem 2.6.3 (Tsukamoto [Ts]) An almost alternating unknot diagram is trivializable by tongue moves, flypes, and crossing number reducing Reidemeister I and II moves.

[6] We confess to going here against established knot theory terminology with a different meaning. Exculpatorily, these classical sort of tunnels will never be treated here.

Downloaded by [CRCNetbase T&F Offices] at 03:10 10 August 2016

Taylor and Francis
Not for Distribution

© 2016 by Taylor & Francis Group, LLC

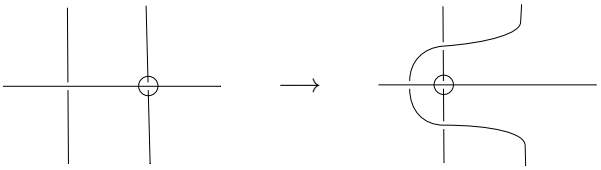


FIGURE 2.8: A tongue move between two almost alternating diagrams. The encircled crossing is to be switched to obtain an alternating diagram.

A tongue move is a special type of wave move, preserving the property almost alternating. In that sense Theorems 1.5.1 and 5.5.4 can be thought of as extending Tsukamoto's result in special diagrams.

Downloaded by [CRCNetbase T&F Offices] at 03:10 10 August 2016

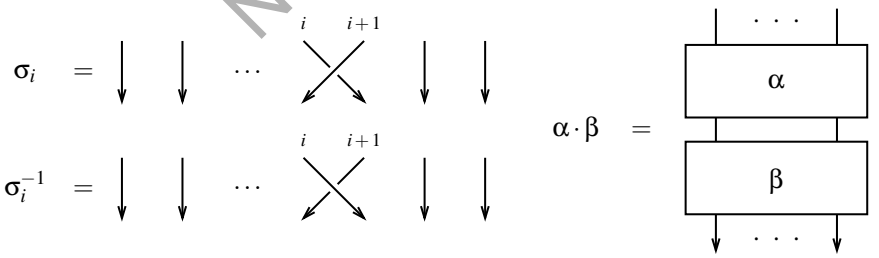
Taylor and Francis
Not for Distribution

2.7 Braids and braid representations

For braid-related knot theory, the basic reference remains Birman's book [Bi2].

The *braid groups* B_n were introduced in the 1930s in the work of Artin [Ar]. An element $\beta \in B_n$ is called an *n-braid*. The braid group B_n on n strands (or strings) is considered to be generated by the Artin *standard generators* σ_i for $i = 1, \ldots, n-1$. These are subject to relations of the type $[\sigma_i, \sigma_j] = 1$ for $|i-j| > 1$, which we call *commutativity relations* (the bracket denotes the commutator) and $\sigma_{i+1}\sigma_i\sigma_{i+1} = \sigma_i\sigma_{i+1}\sigma_i$, which we call *Yang-Baxter* (or shortly YB) relations.

There is a graphical representation of braids where in σ_i strands i and $i+1$ cross, and multiplication is given by stacking.

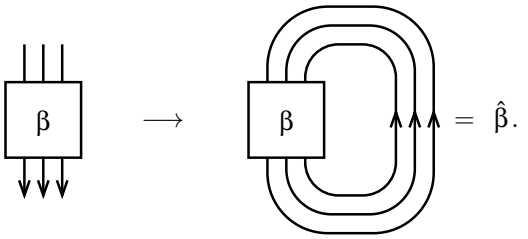


(Here we number strands from left to right and orient them downward. This convention, however, will not matter much.)

Alexander [Al2] related braids to links in real 3-dimensional space by means of a *closure* operation ˆ. In that realm, in became important to understand the *braid representations* of a given link L.

The *closure* $\hat{\beta}$ of a braid β is a knot or link (with orientation):

© 2016 by Taylor & Francis Group, LLC



The *braid index* $b(L)$ of a link L is the smallest number of strands of a braid β whose closure $\hat{\beta}$ is L. See [Mo, FW, Mu4]. Alexander's theorem asserts that $L = \hat{\beta}$ for some braid β.

2.8 Link polynomials

2.8.1 Generalities

The purpose of this section is to give an introduction to link polynomials and their basic properties.

First, let us clarify the meaning of a "link polynomial." It is understood as an invariant of links

$$\{\text{knots and links}\} \to \text{some ring of polynomials}.$$

This means that every link now has its own polynomial. The ring of polynomials will have one or two variables, but we will assume that variables can occur in negative powers, i.e., consider *Laurent* polynomials. We will also allow half-integral powers for links. Coefficients will usually be integers.

The names of the variables will be specified for each invariant. Regarding the polynomials P and F we will introduce shortly, we mention already in advance that there are not only several variable names, but also *several other variable conventions*, differing from each other by possible inversion and/or multiplication of a variable by some fourth root of unity. We include some explanation how our conventions are related to those of some (mostly standard) literature, but the reader should be very careful about this point with comparisons to external sources.

Often variables of polynomials will be set to particular (most generally, complex, but most commonly, integer) values. If one fixes the variable so and varies the link, one then has a scalar (complex respectively integer) valued invariant.

In general we will take the liberty (or sloppiness) to alter notation of how the polynomial depends on the link and the variable. The one or the other will be omitted if

Taylor and Francis
Not for Distribution

Downloaded by [CRCNetbase T&F Offices] at 03:10 10 August 2016

© 2016 by Taylor & Francis Group, LLC

they are suggestive from the context. For example, the Alexander polynomial Δ has the variable t. Thus for each link L, there is a polynomial in $\mathbb{Z}[t^{\pm 1/2}]$, which can be written as $\Delta(L)$, Δ_L, $\Delta(L)(t)$, $\Delta_L(t)$, or $\Delta(t)$, and they all mean the same thing. (There is obviously no danger in confusing a link with a number.) Further, one can replace a link L by a link diagram D on which the polynomial is evaluated.

Thus, we may write the dependence on the link L both as an argument, and as a subscript. However, we will *not* use subscripts for values of the variable: Thus, $\Delta(1)$ will never be written as Δ_1, which will be given a separate meaning later. Let us also agree that when the symbol i is used in an argument, *but not as a subscript*, it always denotes the complex unit $\sqrt{-1}$.

Before we introduce several link polynomials, we need some more notation regarding general polynomials. Let $X \in \mathbb{Z}[t,t^{-1}]$. The *minimal* or *maximal degree* $\min\deg X$ or $\max\deg X$ is the minimal respectively maximal exponent of t with non-zero coefficient in X. Let $\operatorname{span}_t X = \max\deg_t X - \min\deg_t X$, the *span* of *breadth* of X. The coefficient in degree d of t in X is denoted $[X]_{t^d}$ or $[X]_d$. The *leading coefficient* $\max\mathrm{cf}\, X$ of X is its coefficient in degree $\max\deg X$. If $X \in \mathbb{Z}[x_1^{\pm 1}, x_2^{\pm 1}]$, then $\max\deg_{x_1} X$ denotes the maximal degree in x_1. Minimal degree and coefficients are defined similarly, and of course $[X]_{x_1^k}$ is regarded as a polynomial in $x_2^{\pm 1}$.

2.8.2 Skein, Jones, and Alexander polynomial

The *skein polynomial* P [F&, LM, PTr] is a Laurent polynomial in two variables l and m of oriented knots and links and can be defined by being 1 on the unknot and the *skein* relation

$$l^{-1}P\left(\times\right) + lP\left(\times\right) = -mP\left(\;\right)\left(\;\right). \tag{2.9}$$

This equality relates the polynomials of the three fragments in Figure 2.3. We reiterate our understanding from §1.2 that these fragments represent triples of diagrams which are equal except at the depicted spot. We will often denote the diagrams in each triple as in (2.9) (from left to right) by D_+, D_- and D_0.

For a diagram D of a link L, we will use all of the notations $P(D) = P_D = P_D(l,m) = P(L)$ etc. for its skein polynomial, with the self-suggestive meaning of indices and arguments. The convention in (2.9) is similar to (and uses the same variables as) the one in [LM] but differs by the interchange of l and l^{-1}.

The *Jones polynomial* [J] V, and (one variable) *Alexander polynomial* [Al] Δ are obtained from P by the substitutions

$$V(t) = P(-it, i(t^{-1/2} - t^{1/2})), \tag{2.10}$$

$$\Delta(t) = P(i, i(t^{1/2} - t^{-1/2})), \tag{2.11}$$

hence these polynomials also satisfy corresponding skein relations, obtained by the same substitutions in (2.9).

Taylor and Francis
Not for Distribution

Downloaded by [CRCNetbase T&F Offices] at 03:10 10 August 2016

© 2016 by Taylor & Francis Group, LLC

The skein polynomial has at least 8 discoverers (in 5 teams): Freyd–Yetter, Hoste, Lickorish–Millett, Ocneanu, and Przytycki–Traczyk. These discoveries were almost simultaneous, but Przytycki–Traczyk's became known later due to the Marshall Law in Poland. There were attempts from very early on to name the polynomial by some acronym of the discoverers' names (with P and T included or not depending on the origin of the acronym). HOMFLY seems to have imposed itself over LYMFHO, FLYPMOTH, LYMPHTOFU (the latter, according to Bar-Natan [BN3], coined by Rudolph, with U for the Unknown further discoverers), and other imaginable pronounceable permutations of these letters. Acknowledging Przytycki–Traczyk's contribution, it is fairer to write HOMFLY-PT, the possibly best (and most widely used) acronym. There are also other names, like generalized Jones polynomial, Jones–Conway polynomial, 2-variable Jones polynomial, etc. Our choice of "skein polynomial" is pragmatic: it is impartial (and thus fair to all discoverers) and short (typing the 5 small letters of "skein" is a huge relief for one's fingers, as compared to 6 or 8 capital letters, in particular over the length of an entire book). Beware, though, that other authors carelessly assign the name "skein polynomial" also to the other invariants V and Δ which obey skein relations.

The advantage of a skein relation (2.9) is that, when systematically applied, it can recur the value of the skein polynomial of a link diagram to this of simpler diagrams. This gives a method for recursively calculating the polynomial for every link from a diagram and hence also for the Jones and Alexander polynomial.

Instead of Δ, we will sometimes also use the *Conway polynomial* [Co] $\nabla(z)$ with

$$\Delta(t) = \nabla(t^{1/2} - t^{-1/2}). \tag{2.12}$$

Then ∇ satisfies the skein relation

$$\nabla(D_+) - \nabla(D_-) = z\nabla(D_0).$$

It can be easily concluded from this relation that $\nabla(L)(z)$ is a genuine (and not Laurent) polynomial in z. Furthermore, it can be seen that $\nabla(L)(z) \in z^{n(L)-1}\mathbb{Z}[z^2]$; in particular, the Conway polynomial of a knot K satisfies

$$\nabla(K) \in \mathbb{Z}[z^2]. \tag{2.13}$$

Moreover, it is known that for all knots K,

$$[\nabla(K)]_0 = \nabla(0) = \Delta(1) = 1. \tag{2.14}$$

Note that $\max\deg\nabla = 2\max\deg\Delta$, which we use in particular to implicitly restate some of the results of [Cr] in the sequel.

We will need several properties of the Alexander polynomial; here are two classical ones.

Theorem 2.8.1 If L is split, or more generally, bounds a disconnected Seifert surface, then $\Delta_L = 0$. If $\Delta_L \neq 0$ (and in particular L is non-split), then

$$2\max\deg\Delta_L \leq 1 - \chi(L). \tag{2.15}$$

Downloaded by [CRCNetbase T&F Offices] at 03:10 10 August 2016

Taylor and Francis
Not for Distribution

© 2016 by Taylor & Francis Group, LLC

These are properties of the Alexander polynomial that are more obscured than elucidated by skein calculations. Theorem 2.8.1 originates from a connection between the Alexander polynomial and a Seifert surface (which is completely lacking for the skein and Jones polynomial). See, e.g., [Ro]. Kauffman's book [Kf3] treats (2.15) more extensively. We remark that, in Seifert-matrix-related algebraic topology, the Alexander polynomial usually emerges only up to units in $\mathbb{Z}[t,t^{-1}]$; the normalization coming from (2.11) is so that, for *knots K*,

$$\Delta_K(t) = \Delta_K(1/t) \text{ and } \Delta_K(1) = 1. \tag{2.16}$$

For links L, we have $\Delta_L(t) = (-1)^{n(L)-1}\Delta_L(1/t)$ and $\Delta_L(1) = 0$. There are also multivariable Alexander polynomials (for links) and higher order Alexander polynomials, but these will not be treated here.

Let us note in passing that for braid representations of links, there is yet another framework for calculating the Alexander polynomial from the Burau representation (and more generally the skein polynomial from Hecke algebra representations [J2]). Thus there are at least three different approaches to obtain and study properties of Δ:

- the skein relation,
- the Seifert matrix,
- the Burau matrix.

These connections are truly profound, and it is merely for the focus of the present book that we do not cover the second and third ones here, with [Ro] respectively [Bi2] already designated above as classical sources. A good further treatment of skein theory, from an originator himself, is given in [L].

Returning to (2.15), we notice that Crowell–Murasugi show the following further property for Theorem 2.4.1.

Theorem 2.8.2 (Crowell–Murasugi) If D is a non-split alternating diagram, then

$$2\max\deg\Delta_D \;=\; 1-\chi(D)\,. \tag{2.17}$$

The combination of these two theorems implies Theorem 2.4.1. It also implies:

Corollary 2.8.3 An alternating link L is non-split if and only if an alternating diagram D of L is non-split.

Moreover, it follows that (2.15) is an equality for non-split alternating links. Further, alternating links can be replaced in this discussion by positive links, and (2.17) for positive links is treated in [Cr].

For alternating links, Crowell–Murasugi proved one more property.

Theorem 2.8.4 ([Cw, Mu2]) The coefficients of Δ_K for an alternating knot K alternate in sign, i.e., $[\Delta_K]_i[\Delta_K]_{i+1} \le 0$ for all $i \in \mathbb{Z}$.

Taylor and Francis
Not for Distribution

Downloaded by [CRCNetbase T&F Offices] at 03:10 10 August 2016

© 2016 by Taylor & Francis Group, LLC

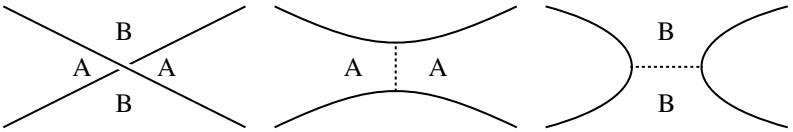
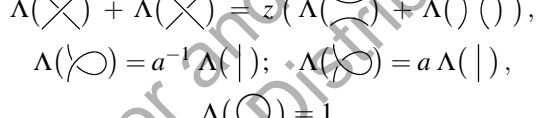


FIGURE 2.9: The A- and B-corners of a crossing, and both of its splittings. The corner A (respectively B) is the one passed by the overcrossing strand when rotated counterclockwise (respectively clockwise) toward the undercrossing strand. A type A (respectively B) splitting is obtained by connecting the A (respectively B) corners of the crossing. It is useful to put a "trace" of each split crossing as an arc connecting the loops at the splitted spot.

2.8.3 Kauffman polynomial and semiadequacy

The *Kauffman polynomial* [Kf] F is usually defined via a regular isotopy invariant $\Lambda(a,z)$ of unoriented links. We use here a slightly different convention for the variables in F, differing from [Kf, Th] by the interchange of a and a^{-1}. Thus in particular, we have for a link diagram D the relation $F(D)(a,z) = a^{w(D)}\Lambda(D)(a,z)$, where $\Lambda(D)$ is the writhe-unnormalized version of the polynomial, given in our convention by the properties



The Jones polynomial V is obtained from F (in our convention) by the substitution (see [Kf2, §III])

$$V(t) = F(-t^{3/4}, t^{1/4} + t^{-1/4}).$$

The *BLMH polynomial* [BLM, Ho] Q is defined by

$$Q(z) = F(1,z).$$

An alternative description of V is given by the Kauffman bracket in [Kf2], which we recall next. A *state* S is a choice of *splittings* of type A or B for any single crossing (see Figure 2.9). Let $\#A(S)$ and $\#B(S)$ denote the number of type A (respectively type B) splittings and $|S|$ the number of (disjoint) circles obtained after all splittings in S. The Kauffman bracket $[D]$ of a diagram D is a Laurent polynomial in a variable A, obtained by summing over all states S of D:

$$[D] = \sum_S A^{\#A(S)-\#B(S)} \left(-A^2 - A^{-2}\right)^{|S|-1}. \tag{2.18}$$

Note that the description of splittings does not require orientation of the diagram.

Downloaded by [CRCNetbase T&F Offices] at 03:10 10 August 2016

Taylor and Francis
Not for Distribution

© 2016 by Taylor & Francis Group, LLC

For oriented diagrams, one of the two splittings of a crossing is the smoothing out; it is the A-splitting if the crossing is positive, and the B-splitting if the crossing is negative.

The Jones polynomial of a link L is related to the Kauffman bracket of some diagram D of L by

$$V_L(t) = \left(-t^{-3/4}\right)^{-w(D)} [D]\Big|_{A=t^{-1/4}}. \tag{2.19}$$

The first major application of the Jones polynomial (proved by Kauffman using the bracket) is the solution of Tait's Crossing conjecture.

Theorem 2.8.5 (Kauffman-Murasugi-Thistlethwaite) If D is a reduced alternating diagram of a link L, then $c(D) = c(L)$.

It will be useful to recall the following (see [St6]).

Theorem 2.8.6 An alternating diagram D represents a positive link L if and only if D is positive (and hence special alternating; recall §2.1).

Corollary 2.8.7 A link is special alternating if and only if it is positive and alternating.

Note that if we replace "link" by "diagram", this is the definition. However, a link may have diagrams with one property and diagrams with another, but no diagram that enjoys both properties simultaneously. The content of the above statement is that this phenomenon does not occur for the properties of (a diagram) being positive and being alternating.

Let the *A-state* $A(D)$ of D be the state where all crossings are A-splitted; similarly, define the B-state.[7] We call a diagram *A-(semi)adequate* if in the A-state no crossing trace (one of the dotted lines in Figure 2.9) connects a loop with itself. We call such a trace a *self-trace*. Similarly, we define B-(semi)adequate. A diagram is *semi-adequate* if it is A- or B-semiadequate, and *adequate* if it is simultaneously A- and B-semiadequate. A link is adequate/semiadequate if it has an adequate/semiadequate diagram. Let us remark here that, while adequacy is only a slight extension of alternation, semiadequacy is a rather wide extension of adequacy. (Semiadequate knots contain completely positive, Montesinos and 3-braid knots.) Semiadequacy is still a fairly general condition, yet it helps settle many technical issues.

For semiadequate knots, the first coefficient of the Jones polynomial is ± 1, almost by definition [LT]. In an attempt to gain more insight into the appearance of the Jones polynomial, in [St5], and [DL, DL2], a detailed study was initiated of some coefficients of V in semiadequate diagrams, gaining an understanding of coefficients 2 and 3. This allowed us to prove, regarding Question 1.3.1:

[7] Note, however, that "$B(D)$" was allocated differently in §2.5.2; we will retain this meaning and never need a symbol for the B-state in the sequel.

Downloaded by [CRCNetbase T&F Offices] at 03:10 10 August 2016

Taylor and Francis
Not for Distribution

© 2016 by Taylor & Francis Group, LLC

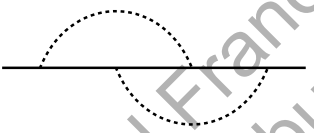
Theorem 2.8.8 ([St11]) For a semiadequate knot (or link) diagram D, we have $V(D) \neq 1$ (or $V(D) \neq (-t^{1/2} - t^{-1/2})^{n(D)-1}$). That is, semiadequate knots and links have non-trivial Jones polynomial.

After the conversion (2.19), we see that the minimal degree of t to which the sum (2.18) contributes occurs in the A-state $A(D)$. For our subsequent arguments, this degree can be more conveniently written for a knot diagram D as

$$m(D) := g(D) - \frac{3}{2}c_-(D) + \frac{s(D) - |A(D)|}{2}. \tag{2.20}$$

If a diagram is A-adequate, only the A-state contributes in degree $m(D)$, and the coefficient is ± 1 (see [LT]). In Bae-Morton [BMo], the contribution of the sum of (2.18) in degree $m(D)$ for general diagrams D was studied. The following simple property will be useful.

Call a self-trace *isolated* if it does not pair up with another self-trace like



Lemma 2.8.9 (Bae-Morton [BMo]) If the A-state of D has an isolated self-trace, then the contribution of the sum of (2.18) in degree $m(D)$ vanishes; that is, $\min\deg V(D) > m(D)$.

Let $F(K) \in \mathbb{Z}[a^{\pm 1}, z]$ be the Kauffman polynomial of a knot K and $Q_K(z) = F_K(1,z)$ be the BLMH polynomial [BLM, Ho]. Let further

$$\begin{aligned} a_-(K) &= \tfrac{1}{2}\max\{m-l : [F(K)]_{a^l z^m} \neq 0\} \quad \text{and} \\ a_+(K) &= \tfrac{1}{2}\max\{m+l : [F(K)]_{a^l z^m} \neq 0\}. \end{aligned} \tag{2.21}$$

It easily follows that $\max\deg_z F(K) \leq a_+(K) + a_-(K)$. Moreover, if the r.h.s. is positive, strict inequality holds because the substitution $F(i,z) = 1$ shows that $\max\mathrm{cf}_z F(K)$ cannot be a single monomial (in a).

In [Th], Thistlethwaite proves that

$$c_\pm(K) \geq a_\pm(K), \tag{2.22}$$

with equality if and only if K is A, respectively, B-semiadequate. Let us say in the following that a (non-strict) inequality of the form '$A \geq B$' is *sharp* or *exact* if $A = B$ and *strict* otherwise (i.e., if $A > B$).

Adequate (in particular alternating) diagrams make both inequalities in (2.22) simultaneously sharp. It follows from these inequalities that for every non-trivial knot K,

$$\max\deg Q(K) \leq \max\deg_z F(K) < c_+(K) + c_-(K). \tag{2.23}$$

Downloaded by [CRCNetbase T&F Offices] at 03:10 10 August 2016

Taylor and Francis
Not for Distribution

© 2016 by Taylor & Francis Group, LLC

(If $c_-(K) = c_+(K) = 0$, then K is simultaneously positive and negative, and it is therefore trivial; see for example [St6], or the below remarks on the signature.)

It follows further that adequate diagrams are of minimal crossing number (thus extending Theorem 2.8.5).

One can also see that adequate diagrams of the same link have the same writhe. Note that for alternating diagrams, this writhe invariance follows from Theorem 2.6.2 because a flype preserves the writhe. It should be emphasized, though, that Theorem 2.6.2 is a very sophisticated tool, and the proof of writhe invariance using (merely) polynomial invariants must be considered to be simpler.

The writhe invariance has a somewhat illustrious history. I am grateful to K. Perko for its clarification. It appears, stated for arbitrary minimal crossing diagrams, as a *theorem* in Little's 1900 paper [Li] (communicated by Tait himself!), and that statement was later cemented by Dehn and Heegaard in their encyclopedia article [DH, p. 212]. This was thus apparently believed to be right for decades until it was surprisingly refuted by the Perko pair [Pe].

Taylor and Francis
Not for Distribution

2.9 MWF inequality, Seifert graph and graph index

In [Mo, FW] it was proved that

$$\mathrm{mwf}(L) := \frac{1}{2}\,\mathrm{span}_l P(L) + 1 \le b(L)\,, \tag{2.24}$$

the *Morton–Williams–Franks* (MWF) *inequality*. Because we will need the left-hand side of this inequality later, let us write it as $\mathrm{mwf}(L)$ and call it the *Morton–Williams–Franks bound* for $b(L)$. The inequality (2.24) results from two other inequalities, due to Morton, namely that for a diagram D, we have

$$1 - s(D) + w(D) \le \min\deg_l P(D) \le \max\deg_l P(D) \le s(D) - 1 + w(D)\,. \tag{2.25}$$

Williams–Franks showed these inequalities for the case of braid representations. Later it was observed from the algorithms of Yamada [Ya] and Vogel [Vo] that the braid version is actually equivalent to—and not just a special case of—the diagram version. (These algorithms allow us to transform any diagram D into a braid diagram without altering $s(D)$ and $w(D)$.) Nonetheless we will refer below to (2.25) as "Morton's inequalities."

These inequalities were later improved in [MP] in a way that helps settling the braid index problem for many links (see Chapter 7 or also [Oh]). For this purpose, Murasugi–Przytycki developed the concept of *index of a graph*. We recall some main points of Murasugi–Przytycki's work, referring to [MP] for further details, and offer caution to the correction explained in [St20] and reviewed here in §7.1–7.3.

Downloaded by [CRCNetbase T&F Offices] at 03:10 10 August 2016

© 2016 by Taylor & Francis Group, LLC

Definition 2.9.1 Let G be a signed graph (each edge carries a sign $+$ or $-$). For a vertex v in G let $G_v = G/v$ be the graph obtained by contracting the star of v, that is, the set of edges of which v is one of the endpoints (§2.5). Let $G \setminus v$ be the graph obtained from G by deleting all of these edges and v.

We say that v is a *cut vertex* if $G \setminus v$ is disconnected. A cut vertex decomposes G into a *block sum* or *join* $G_1 * G_2$ of two graphs $G_{1,2}$. If $G = G_1 * \cdots * G_n$ and all G_i have no cut vertex, the G_i will be called *block components* or *join factors* of G.

Definition 2.9.2 We define (recursively) a sequence of edges $\mu = (e_1, \dots, e_n)$ to be *independent* in a graph G if the following conditions are satisfied.

1. The empty (edge) sequence is independent per definition.
2. Let e_1 connect vertices $v_{1,2}$. Then we demand that e_1 is *simple*, i.e., there is no other edge connecting $v_{1,2}$, and that $e_2, \dots, e_n$ is independent in (one of) G_{v_1} or G_{v_2} (i.p., $e_2, \dots, e_n$ are disjoint from the set of edges incident to v_1 or v_2 respectively).

An *independent set* is a set of edges admitting an ordering as an independent sequence.

The *index* $\mathrm{ind}\,(G)$, respectively *positive index* $\mathrm{ind}_+(G)$ and *negative index* $\mathrm{ind}_-(G)$ of G are defined as the maximal size/length of an independent edge set (or sequence), respectively independent positive or negative edge set/sequence in G. An independent set/sequence is *maximal independent* if it realizes the index of G.

(This index is to be kept strictly apart from the index of a Seifert circle of §2.1. Both are not unrelated, but we do not investigate their relation and use them in quite separate contexts.)

Now to each link diagram D we associate its *Seifert graph* $G = \Gamma(D)$, which is a planar bipartite signed graph. It consists of a vertex for each Seifert circle in D and an edge for each crossing, connecting two Seifert circles. Each edge is signed by the skein sign of the crossing it represents. We will for convenience sometimes identify crossings/Seifert circles of D with edges/vertices of G. Let also $\mathrm{ind}_{(\pm)}(D) = \mathrm{ind}_{(\pm)}(\Gamma(D))$.

Proposition 2.9.3 (see [MP, (8.4) and (8.8)]) If D is a diagram of an oriented link L, then

$$\max \deg_l P(L) \leq w(D) + s(D) - 1 - 2\,\mathrm{ind}_+(D) \tag{2.26}$$

$$\min \deg_l P(L) \geq w(D) - s(D) + 1 + 2\,\mathrm{ind}_-(D) \tag{2.27}$$

$$b(L) \leq \mathrm{mpb}\,(D) := s(D) - \mathrm{ind}\,(D)\,. \tag{2.28}$$

For any diagram D, we have

$$\mathrm{ind}_+(D) + \mathrm{ind}_-(D) \geq \mathrm{ind}\,(D)\,. \tag{2.29}$$

Downloaded by [CRCNetbase T&F Offices] at 03:10 10 August 2016

Taylor and Francis
Not for Distribution

© 2016 by Taylor & Francis Group, LLC

For alternating (and more generally homogeneous [Cr]) diagrams D, equality holds because each join factor of $\Gamma(D)$ contains only edges of the same sign. This implies that if for such diagrams (2.26), (2.27) are sharp, then (2.24) and (2.28) become sharp, too.

It is possible to partially reconstruct D from $\Gamma(D)$ and completely when D is special and $\Gamma(D)$ is given in a planar embedding. If $\Gamma(D)$ has block components (Definition 2.9.1) G_i, then we call the (special) diagrams D_i with $\Gamma(D_i) = G_i$ the *Murasugi atoms* of D. Note that each separating Seifert circle of D gives a cut vertex of $\Gamma(D)$. The join operation of G_i at these cut vertices corresponds to an operation of identifying two Seifert circles in D_i, which is called planar (or diagrammatic) *Murasugi sum* (or *-product; see, e.g., [MP2]). Thus an alternative way to specify the Murasugi atoms D_i is to say that they are the prime factors of the Murasugi summands of D (see also Definition 3.2.1 below, and [QW]). However, the information of how under a Murasugi sum in D crossings attached to a Seifert circle on either side are arranged relatively to each other is lost in $\Gamma(D)$.

The following theorem of Murasugi–Przytycki is important:

Theorem 2.9.4 (Theorem 2.5 in [MP]) The index is additive under block sum of bipartite graphs, i.e., $\operatorname{ind}(G_1 * G_2) = \operatorname{ind}(G_1) + \operatorname{ind}(G_2)$ if $G_{1,2}$ are bipartite.

We will treat below, simultaneously with (the sharpness of) MWF, the following conjecture of theirs.

Conjecture 2.9.5 (Murasugi–Przytycki) If D is an alternating diagram of a link L, then $b(L) = \operatorname{mpb}(D)$.

Remark 2.9.6 We observed that for a special diagram D, Seifert circles coincide with the boundary of a collection of regions. It is easy to see that these regions do not share an edge, and thus they are the regions receiving the same color in a checkerboard coloring of D. This establishes that *for a special diagram D, the Seifert graph* $\Gamma(D)$ *coincides with one of the checkerboard graphs of D* (see §2.5.2).

2.10 The signature

The *signature* σ is a $\mathbb{Z}$-valued invariant of knots and links. Originally it was defined in terms of Seifert matrices [Ro]. We have that $\sigma(L)$ has the opposite parity to the number of components of a link L, whenever the *determinant*

$$\det(L) := \left|\Delta_L(-1)\right| \tag{2.30}$$

Downloaded by [CRCNetbase T&F Offices] at 03:10 10 August 2016

Taylor and Francis
Not for Distribution

© 2016 by Taylor & Francis Group, LLC

does not vanish,

$$\det(L) \neq 0\,. \tag{2.31}$$

Note that (2.31) always occurs for L being a knot. We have from (2.12), (2.13) and (2.14)

$$\Delta_K(-1) = \nabla_K(2i) \equiv 1 \bmod 4\,. \tag{2.32}$$

In particular, $\det(K)$ is always odd. Thus σ takes only even values on knots.

Most of the early work on the signature was done by Murasugi [Mu5], who showed several properties of this invariant.

For links $L_{\pm,0}$ with diagrams as in (2.9), we have

$$\sigma(L_+) - \sigma(L_-) \quad \in \quad \{0,1,2\} \tag{2.33}$$

$$\sigma(L_\pm) - \sigma(L_0) \quad \in \quad \{-1,0,1\}\,. \tag{2.34}$$

(Note: In the first property one can also have $\{0,-1,-2\}$ instead of $\{0,1,2\}$ because other authors, like Murasugi, take σ to be with opposite sign. Thus (2.33) not only defines a property, but it also specifies our sign convention for σ.)

Next, σ obeys additivity under split union (denoted by '$\sqcup$') and connected sum (denoted by '#'):

$$\sigma(L_1 \# L_2) = \sigma(L_1 \sqcup L_2) = \sigma(L_1) + \sigma(L_2)\,. \tag{2.35}$$

The following other property is very useful: $\sigma(!L) = -\sigma(L)$, where $!L$ is the mirror image of L.

Further, Murasugi found the following important relation between $\sigma(K)$ and $\det(K)$ for a *knot* K.

$$\begin{aligned} \sigma(K) \equiv 0\,(4) \quad &\Longleftrightarrow \quad \det(K) \equiv 1\,(4) \\ \sigma(K) \equiv 2\,(4) \quad &\Longleftrightarrow \quad \det(K) \equiv 3\,(4) \end{aligned} \tag{2.36}$$

Furthermore,

$$\text{if } \det = 1, \text{ then even } 8 \mid \sigma \tag{2.37}$$

because of the property of signatures of unimodular quadratic forms.

Because of (2.32), one can modify (2.36) to

$$\begin{aligned} \sigma(K) \equiv 0\,(4) \quad &\Longleftrightarrow \quad \Delta_K(-1) > 0 \\ \sigma(K) \equiv 2\,(4) \quad &\Longleftrightarrow \quad \Delta_K(-1) < 0\,. \end{aligned} \tag{2.38}$$

Note further that for an *alternating* knot K of genus $g = g(K)$, because of (2.17), we have

$$\max \mathrm{cf}\,\Delta_K = \max \mathrm{cf}\,\nabla_K = [\Delta_K]_g = (-1)^g [\nabla_K]_{2g}$$

and because of Theorem 2.8.4,

$$\mathrm{sgn}(\Delta_K(-1)) = (-1)^g \mathrm{sgn}([\Delta_K]_g)\,.$$

Taylor and Francis
Not for Distribution

Downloaded by [CRCNetbase T&F Offices] at 03:10 10 August 2016

© 2016 by Taylor & Francis Group, LLC

In combination with (2.38), this means that g and $\sigma \bmod 4$ determine the sign of $\max\mathrm{cf}\,\nabla_K = [\Delta]_{\pm g}$ (when Δ is normalized as in (2.16)):

$$\mathrm{sgn}([\Delta]_{\pm g}) = (-1)^{g+\sigma/2}. \tag{2.39}$$

We will return to this property toward the end of our long treatise.

A further use of (2.36) is that these conditions, together with the initial value

$$\sigma(\bigcirc) = 0$$

for the unknot and shortcuts like (2.35), allow one to calculate σ for most links, incl. all knots.

If K is a positive knot, then it was widely observed (for the first time probably in [CG]) that $\sigma(K) > 0$ (and hence $\sigma(K) \geq 2$). Przytycki noticed that signatures of positive links can be studied based on Taniyama's previous work [Ta], but their manuscript [PT], despite being announced long ago, was only recently completed. In particular, $\sigma(K) \geq 4$ if $g(K) \geq 2$; also $\sigma(K) > 0$ if K is almost positive or 2-almost positive, except a twist knot. See [St3, St7, St2] for previously published (though not previously written, and not identical) proofs (and Chapter 6 below for more discussion).

There is a nice combinatorial formula for σ for alternating diagrams. We use this formula here to remind us of the below property of signatures for alternating links.

Proposition 2.10.1 Let L be a non-split alternating link. Then

$$\sigma(L) \leq 1 - \chi(L), \tag{2.40}$$

and equality holds if and only if L is special alternating and positive.

Here we emphasize the equality case far more than (2.40) itself. This inequality is true for much more general links, including all links with connected minimal genus Seifert surfaces, in particular also all non-split positive links and links with non-zero Alexander polynomial (recall Theorem 2.8.1). For knots (where always $\Delta \neq 0$), (2.40) follows as a combination of (2.15) and Theorem 9.1.3. (There are ways to extend this reasoning to links, but they definitely lead too far afield here.)

Proposition 2.10.1 is well known but will be relevant in the following, and because I am not certain where to find a proof, I include one.

Proof. This inequality can be proved using checkerboard colorings (§2.5.2). We consider now only the case that D is alternating. Then it is well known that all A-corners of all crossings of D according to Figure 2.9 arise as corners of regions of one color, and all B-corners for the regions of the other color. Our convention is that the white regions contain all A-corners.

By Corollary 2.8.3, D is non-split. By Corollary 2.8.7, we need to show that equality in (2.40) occurs if and only if D is positive.

Downloaded by [CRCNetbase T&F Offices] at 03:10 10 August 2016

Taylor and Francis
Not for Distribution

© 2016 by Taylor & Francis Group, LLC

We will use the formula for σ on p. 437 of [Kf] and negate the r.h.s. (we want the positive trefoil to have positive signature; the convention there is opposite). We get, using (2.7),

$$\sigma(D) = \frac{w(D)}{2} + \frac{c(D) - 2(W(D) - 1)}{2}. \tag{2.41}$$

Then with $w(D) = c(D) - 2c_-(D)$, (2.40) transcribes into

$$W(D) \leq s(D) + c_-(D). \tag{2.42}$$

We will see this inequality (together with its equality case) by induction on the number of blocks or Murasugi atoms (see definition in §2.9).

To start induction, assume D is special. Since D is non-split and alternating, we know that D is positive or negative. If D is positive, then $c_-(D) = 0$, and $W(D) = s(D)$, and we have (2.42) with equality. If D is negative, then $B(D) = s(D)$ and $c_-(D) = c(D)$. Thus (2.42) becomes, in a strict form,

$$c(D) + 2 - s(D) < c(D) + s(D),$$

which is true for $s(D) > 1$. (When $s(D) = 1$, then D is a circle.)

To complete induction, it suffices to observe how the quantities in (2.42) behave under Murasugi sum $*$. We have for $D = D_1 * D_2$,

$$\begin{aligned} W(D) &= W(D_1) + W(D_2) - 1, \\ s(D) &= s(D_1) + s(D_2) - 1, \\ c_-(D) &= c_-(D_1) + c_-(D_2). \end{aligned}$$

For the first equality, note that under Murasugi sum, one region of D_1 and one region of D_2 are annihilated, and exactly one is black and one is white. This shows (2.42) (including its equality case) and proves Proposition 2.10.1. □

2.11 Genus generators

Now let us recall, from [St4, St2], some basic facts concerning knot generators of a given genus. An explanation is also given in section 5.3 in [Cr2]. (There are several equivalent forms of these definitions, and we choose here one that closely leans on the terminology of Gauss diagrams; see for example [St].) We will also set up some notations and conventions used below. The situation for links is discussed only briefly here and in much more detail in Chapter 3.

Definition 2.11.1 We call two crossings p, q in a knot diagram *linked*, and write $p \cap q$ if passing their crossing points along the orientation of D we have the cyclic order $pqpq$ and not $ppqq$.

Downloaded by [CRCNetbase T&F Offices] at 03:10 10 August 2016

Taylor and Francis
Not for Distribution

© 2016 by Taylor & Francis Group, LLC

Definition 2.11.2 Let D be a knot diagram, and p and q be crossings.

(i) We call p and q *(twist) equivalent*, $q \simeq p$, if for all $r \neq p,q$ we have $r \cap p \iff r \cap q$.

(ii) We call p and q *$\sim$-equivalent* and write $p \sim q$ if p and q are equivalent and $p \not\cap q$.

(iii) Similarly, p and q are called *$\overset{\sim}{*}$-equivalent*, $p \overset{\sim}{*} q$, if p and q are equivalent and $p \cap q$.

(iv) Finally, call two crossings p and q *Seifert equivalent* if they connect the same two Seifert circles.

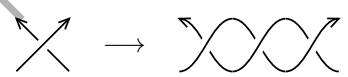
Let us record the following easy but useful observation.

Lemma 2.11.3 $\overset{\sim}{*}$-equivalent crossings are Seifert equivalent. The converse is true in special diagrams. □

Definition 2.11.4 A $\sim$-equivalence class consisting of one crossing is called *trivial* and a class of more than one crossing *non-trivial*. A $\sim$-equivalence class is *reduced* if it has at most two crossings; otherwise it is *non-reduced*.

Definition 2.11.5 Let $t(D)$ be the number of $\sim$-equivalence classes of D. For $i = 1,\dots,t(D)$ let $t_i(D)$ be the number of crossings in the i-th $\sim$-equivalence class. Then for $i = 1,\dots,t(D)$ and $j = 1,\dots,t_i(D)$ let $p(D,i,j)$ be the j-th crossing in the i-th $\sim$-equivalence class of D.

Definition 2.11.6 A $\vec{t}_2'$ *move* or *twist* is given up to mirroring by



(2.43)

We will call it positive or negative, depending on whether it acts on a crossing of the according sign.

Thus a $\vec{t}_2'$ move at a crossing x in a diagram D creates a pair of $\sim$-equivalent crossings to x. (This property specifies the $\vec{t}_2'$ move alternatively up to flypes.)

Definition 2.11.7 An alternating diagram D is called $\vec{t}_2'$ *irreducible* or *generating diagram* if all $\sim$-equivalence classes are reduced, that is, $t_i(D) \leq 2$ for $i = 1,\dots,t(D)$. An alternating knot K is called a *generator* if some of its alternating diagrams is generating. The diagrams obtained from D by $\vec{t}_2'$ moves and crossing changes form the *sequence* or *series* $\langle D \rangle$ of D.

Downloaded by [CRCNetbase T&F Offices] at 03:10 10 August 2016

Taylor and Francis
Not for Distribution

© 2016 by Taylor & Francis Group, LLC

(In many cases it will be useful to restrict oneself in $\langle D \rangle$ to alternating or positive diagrams.)

Observe that, because $\sim$-equivalence is invariant under flypes, Theorem 2.6.2 implies that some alternating diagram of K is generating if and only if all its alternating diagrams are such.

A flype in Figure 1.2 is called *trivial* if one of the tangles P, Q contains only crossings equivalent to the crossing admitting the flype. A flype is of type A if the strand orientation is so that strands on the left/right side of each tangle are directed equally w.r.t. the tangle (i.e., both enter or both exit). Otherwise it is a flype of type B. Compare Figure 2.6. So the property of a crossing to admit a type A respectively type B flype is an invariant of the $\overset{*}{\sim}$ respectively $\sim$-equivalence class.

Theorem 2.11.8 ([St4]) There exist only finitely many generators of a given genus. All diagrams of that genus can be obtained from diagrams of these generators under $\vec{r}_2'$ twists, flypes, and crossing changes.

It may be worth noticing that in this sequence of operations, all flypes can be performed at the beginning.

The finiteness of generators, together with the Flyping theorem [MT], shows the following theorem.

Theorem 2.11.9 (see [St4]) Let $a_{n,g}$ be the number of prime alternating knots K of genus $g(K) = g$ and crossing number $c(K) = n$. Then for $g \geq 1$

$$\sum_n a_{n,g} x^n = \frac{R_g(x)}{(x^{p_g} - 1)^{d_g}},$$

for some polynomial $R_g \in \mathbb{Q}[x]$, and $p_g, d_g \in \mathbb{N}$.

Alternatively, this statement can also be written in the following form: There are numbers p_g (period), n_g (initial number of exceptions) and polynomials $P_{g,1}, \ldots, P_{g,p_g} \in \mathbb{Q}[n]$ with

$$a_{n,g} = P_{g,n \bmod p_g}(n) \text{ for } n \geq n_g\,. \tag{2.44}$$

This was explained roughly in [St4] and then in more detail in [SV], where we made effort to characterize the leading coefficient of these polynomials $P_{g,i}$. Even if the polynomials vary with a very large period p_g, the leading coefficients depend only on the parity of n (see §10.3). The degrees of all $P_{g,i}$ are also the same and equal to 1 less than the maximal number of $\sim$-equivalence classes of diagrams of canonical genus g, with the exception $g = 1$, in which case this degree is 1 or 2 depending on whether i is even or odd.

In practice (in particular as we will see below), it is important to obtain the list of generators for small genus. Genus 1 is easy and was also observed independently.

Downloaded by [CRCNetbase T&F Offices] at 03:10 10 August 2016

Taylor and Francis
Not for Distribution

© 2016 by Taylor & Francis Group, LLC

Theorem 2.11.10 ([St4]; see also [Ru]) There are two generators of genus 1, the trefoil and figure-8-knot.

Genus 2 and 3 require much more work. For reasons already suggested, it is sufficient to find prime diagrams, and by Theorem 2.1.1, prime generators.

Theorem 2.11.11 ([St2]) There are 24 prime generators of genus 2, which are 5_1, 6_2, 6_3, 7_5, 7_6, 7_7, 8_{12}, 8_{14}, 8_{15}, 9_{23}, 9_{25}, 9_{38}, 9_{39}, 9_{41}, 10_{58}, 10_{97}, 10_{101}, 10_{120}, 11_{123}, 11_{148}, 11_{329}, 12_{1097}, 12_{1202}, and 13_{4233}, and 4017 prime generators of genus 3.

A classification (by means of obtaining the list of prime generators) for genus $g = 4$ is also possible and will be explained below, along with a general statement in Theorem 3.1.1.

It follows from Theorem 2.6.2 that the series of different (alternating) diagrams of the same generating knot are equivalent *up to mutations*. These mutations arise when a $\bar{t}_2'$ move is performed at a crossing admitting a type A flype. When using tests involving the polynomial invariants, which do not change under mutations, it is legitimate that *a priori* we fix *a single specific diagram D for each generator K* and work only with (the series of) this diagram. Then we write $t(K) = t(D)$, $t_i(K) = t_i(D)$ and $p(K,i,j) = p(D,i,j)$ and speak of the sequence or series of K. With some simple observations, we will be able to comfort ourselves with this assumption in Chapter 7, whereas in Chapter 5 more care is needed.

Further, we will often discard diagrams with crossings of different sign within the same $\sim$-equivalence class. Such diagrams have a trivial clasp after flypes and are usually of little interest. We will call a $\sim$-equivalence class *positive* or *negative* depending on the sign of its crossings. There is an important exception to this rule, which occurs for $t_i = 2$: Then we do not cancel one positive and one negative crossing (because we need to keep the $\sim$-equivalence class).

Turning to *link* diagrams, analogous, although slightly modified, definitions are possible. We call alternating diagrams *generating* or $\bar{t}_2'$*-irreducible* if they cannot be reduced by flypes and inverses of the move (2.43). Definition 2.11.7 of a *series* can be restated here.

The finiteness result in Theorem 2.11.8 can be extended to link diagrams of a given genus and a given number of components. We will discuss this in detail in Theorem 3.1.1.

For link diagrams, we can define two crossings as $\sim$*-equivalent* if after a sequence of flypes they can be made to form a *reverse clasp*  ; it is an exercise to check that this is an equivalence relation (see [St2]). Similarly, we call two crossings $\tilde{*}$*-equivalent* if after a sequence of flypes they can be made to form a *parallel clasp*  . We observed (see [St4, St2, STV]) that $\sim$- and $\tilde{*}$-equivalent crossings of a knot diagram are linked with the same set of other crossings in the diagram.

Downloaded by [CRCNetbase T&F Offices] at 03:10 10 August 2016

Taylor and Francis
Not for Distribution

© 2016 by Taylor & Francis Group, LLC

Thus, the present definition is a genuine extension of the previous one for the case of knots.

In relation to clasps, let us stipulate that the clasp is *positive*, *negative*, or *trivial/resolved*, depending on the sign of its crossings. Removing a trivial clasp by a Reidemeister II move, after possibly switching a crossing in a non-trivial clasp, will be called *resolving* the clasp.

2.12 Knots vs. links

To conclude our preparation, let us make more precise the separation between knots and links in the following work.

The methods developed in this text put into prospect to adapt results about "k-almost positive knots" and "alternating knots of genus k" to statements that apply to "$\tilde{k}$-almost positive l-component links" and "alternating l-component links of genus $\tilde{k}$" for $\tilde{k}+l \leq k+1$.

One main technical reason for focusing on knots is that our computational tools are heavily based on knot diagrams (see similarly Remark 5.4.5). The results where links are covered appear in Chapter 3 (here the inclusion of links is essential and will be used elsewhere) and in §7.5. They rely on purely theoretical arguments, and computation is not needed. To some extent, we managed to adapt computations to the link case, as shown in Theorem 9.2.7. Its proof should serve to explain how to extend arguments to links in other parts as well, however, more effort will be needed. We also expect that some statements for links would look less pleasant than for knots.

Taylor and Francis
Not for Distribution

Downloaded by [CRCNetbase T&F Offices] at 03:10 10 August 2016

© 2016 by Taylor & Francis Group, LLC